

A Bit of Algebra

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Date: August 16, 2026

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Category Theory I

1.1 Grothendieck Universe and Categories

Grothendieck Universes

Before formally discussing category theory, we need to address a minor detail. Although it is not crucial for actual learning, filling in some logical gaps can always provide a sense of reassurance. Let us first give the conventional definition of a category as it appears in current textbooks. Once again, we reiterate that our study of basic algebra in these notes is still based on the **ZFC** axiom system.

1.1.1 DEFINITION. A *category* $\mathbf{C}$ consists of:

1. a set $\text{Ob}(\mathbf{C})$, whose elements are called the *objects* of $\mathbf{C}$;
2. a set $\text{Mor}(\mathbf{C})$, whose elements are called the *morphisms* of $\mathbf{C}$, together with a pair of functions $\text{Mor}(\mathbf{C}) \xrightarrow{s} \text{Ob}(\mathbf{C}), \text{Mor}(\mathbf{C}) \xrightarrow{t} \text{Ob}(\mathbf{C})$ where s and t give the *source* and *target* of a morphism, respectively. For $X, Y \in \text{Ob}(\mathbf{C})$, we denote $\text{Hom}_{\mathbf{C}}(X, Y) = \{s^{-1}(X) \cap t^{-1}(Y)\}$ called the *Hom-set*, its elements are called morphisms from X to Y ;
3. for each object X , a designated element $\text{id}_X \in \text{Hom}_{\mathbf{C}}(X, X)$ called the *identity morphism* from X to itself;
4. for any objects $X, Y, Z \in \text{Ob}(\mathbf{C})$, a *composition* function

$$\begin{aligned} \circ : \text{Hom}_{\mathbf{C}}(Y, Z) \times \text{Hom}_{\mathbf{C}}(X, Y) &\rightarrow \text{Hom}_{\mathbf{C}}(X, Z) \\ (f, g) &\mapsto f \circ g \end{aligned}$$

(often written simply as fg when no confusion arises). This composition must satisfy these two conditions:

- (a) (Associativity): for any morphisms $h, g, f \in \text{Mor}(\mathbf{C})$ such that the composites $f \circ (g \circ h)$ and $(f \circ g) \circ h$ are defined $f \circ (g \circ h) = (f \circ g) \circ h$. Hence both sides can be written as $f \circ g \circ h$ or fgh ;
- (b) (Identity Law): for any morphism $f \in \text{Hom}_{\mathbf{C}}(X, Y)$, $f \circ \text{id}_X = f = \text{id}_Y \circ f$.

When we discuss categories, we inevitably speak of “the category of all $\dots$ ”. For example, the category of all sets, the category of all groups, and even the category of all categories. Specifically, when we discuss categories, we are faced with two classes: $\text{Ob}(\mathbf{C})$ and $\text{Mor}(\mathbf{C})$. When we speak of the category of all sets, $\text{Ob}(\mathbf{C})$ is the class of all sets, which is illegal within the **ZFC** axiomatic system — to be precise, it is not logically well-formed. As discussed in the previous section, we must be very careful with such phrasing — at least within the **ZFC** axiomatic system, we cannot formulate it that way. Grothendieck proposed a method to allow us to discuss categories “safely” within the **ZFC** framework, namely the Grothendieck universe.

1.1.2 DEFINITION. A *Grothendieck universe* is a nonempty set $\mathbf{U}$ with properties:

1. $x \in u \in \mathbf{U} \rightarrow x \in \mathbf{U}$;
2. $u, v \in \mathbf{U} \rightarrow \{u, v\} \in \mathbf{U}$;
3. $u \in \mathbf{U} \rightarrow \mathcal{P}(u) \in \mathbf{U}$;
4. If $I \in \mathbf{U}$ and $\{x_i : i \in I\}$ is a set of elements of $\mathbf{U}$, then $\bigcup \{x_i : i \in I\} \in \mathbf{U}$;

There are two simple examples of Grothendieck universes: the empty set and the set of all hereditarily finite sets $\mathbf{V}_\omega$. We call the $\mathbf{V}_0$ and $\mathbf{V}_\omega$ the trivial universes. Other examples are more difficult to construct.

1.1.1 PROPOSITION. *The following properties are clear:*

1. If $x \in \mathbf{U}$, then $\{x\} \in \mathbf{U}$;
2. If $x \subset y$, where $y \in \mathbf{U}$, then $x \in \mathbf{U}$, in particular $\emptyset \in \mathbf{U}$;
3. If $x, y \in \mathbf{U}$, then $(x, y) = \{x, \{x, y\}\} \in \mathbf{U}$;
4. If $x, y \in \mathbf{U}$, then $x \cup y$ and $x \times y$ are elements of $\mathbf{U}$;
5. If $I \in \mathbf{U}$ and $\{x_i\}_{i \in I}$ is a family of elements of $\mathbf{U}$, then the set $\prod_{i \in I} x_i \in \mathbf{U}$;
6. If $x \in \mathbf{U}$, then $x \cup \{x\} \in \mathbf{U}$ and therefore $\mathbf{N} \subset \mathbf{U}$;
7. If $x \in \mathbf{U}$, then $\bigcup x \in \mathbf{U}$.

Clearly, if $\mathbf{U}$ is an universe, then we can start from the elements of this universe and perform almost all set-theoretic operations while ensuring that we never leave this universe. More precisely, the **ZFC** axioms strictly limit the ways in which new sets can be produced from known sets, but there is one exception: the Axiom of Replacement, which can construct a set outside this universe. To address this issue, Grothendieck propose the following axiom:

AXIOM (UA). Every set is an element of some universe.

He suggested to use **ZFC** together with requirement that every set X be contained in some Grothendieck universe guaranteeing at least an infinite sequence of Grothendieck universes: $\mathbf{U}_0 \in \mathbf{U}_1 \in \dots \in \mathbf{U}_n \in \mathbf{U}_{n+1} \in \dots$. As Grothendieck universes are transitive such a sequence is also cumulative in the sense that

$$\mathbf{U}_0 \subset \mathbf{U}_1 \subset \dots \subset \mathbf{U}_n \subset \mathbf{U}_{n+1} \subset \dots$$

1.1.3 DEFINITION. An **U-category** is a category whose objects and morphisms form **U**-classes, and whose morphisms sets are **U**-classes, i.e., $\text{Ob}(\mathbf{C})$ and $\text{Mor}(\mathbf{C})$ are **U**-classes, and $\text{Hom}_{\mathbf{C}}(X, Y)$ is a **U**-set for any $X, Y \in \text{Ob}(\mathbf{C})$.

Clearly, the category of all **U**-sets is a **U**-category: since $\mathbf{U} \subset \mathbf{U}$, the object class is a **U**-class. Every morphism $f : X \rightarrow Y$ with $X, Y \in \mathbf{U}$ is a subset of $X \times Y$; since $X \times Y \in \mathbf{U}$ by universe closure and any subset of an element of $\mathbf{U}$ is again an element of $\mathbf{U}$ (proposition 1.(2)), each morphism lies in $\mathbf{U}$, so the class of all morphisms is also a subset of $\mathbf{U}$ and therefore a **U**-class. Finally, since $X \times Y \in \mathbf{U}$ and $\mathcal{P}(X \times Y) \in Y$, by proposition 1.(2) we obtain that the set $\text{Hom}_{\mathbf{C}}(X, Y)$ of all functions from X to Y is indeed an element of $\mathbf{U}$, hence a **U**-set. We denote this category by $\text{Set}_{\mathbf{U}}$.

On the other hand, if we want to discuss a larger category, such as the category of all **U**-classes, then since the cardinality of $\mathcal{P}(U)$ is strictly greater than that of $\mathbf{U}$, its object set is not a **U**-class. What should we do then? By making reasonable use of the Universe Axiom (UA), we can take a universe $\mathbf{U}'$ that is larger than $\mathbf{U}$. It is easy to verify that the original **U**-classes become $\mathbf{U}'$ -sets in this new universe, and naturally all **U**-classes form a $\mathbf{U}'$ -set, hence they form a $\mathbf{U}'$ -class. This resolves the issue. We will not discuss the remaining details for now, such as the definition of subcategories, etc. Similarly, if we wish to construct even larger categories, we can always find a larger universe in this manner.

1.1.2 THEOREM. *The following two axioms are equivalent:*

1. **UA**: for every set x , there is an universe $\mathbf{U}$ such that $x \in \mathbf{U}$.
2. **BC**: for every cardinal κ , there is a strongly inaccessible cardinal $\lambda > \kappa$.

Proof. The proof of this theorem is not of great significance in these notes. Interested readers may consult Wikipedia or other relevant sources. $\square$

In fact, any non-trivial Grothendieck universe is of the form $\mathbf{V}_\kappa$ where κ is the strongly inaccessible cardinal. Since the existence of strongly inaccessible cardinals cannot be proved from the axioms of **ZFC**, the existence of universes other than the empty set and $\mathbf{V}_\omega$ cannot be proved from **ZFC** either.

The properties of strongly inaccessible cardinals tell us that starting from elements of $\mathbf{V}_\kappa$, no operation within the **ZFC** system can go beyond V_κ . If the reader has some acquaintance with mathematical logic, a more precise statement can be made: for a strongly inaccessible cardinal κ , the universe $\mathbf{U} := \mathbf{V}_\kappa$ together with the membership relation $\in$ constitutes a model of **ZFC**. This is tantamount to running a copy of set theory virtually inside set theory. One may view it as follows: if the elements of $\mathbf{V}_\kappa$ are regarded as “sets”, then from the perspective of the model $(V_\kappa, \in)$, the “classes” are precisely the elements of $\mathbf{V}_{\kappa+1} = \mathcal{P}(\mathbf{V}_\kappa)$.

1.1.4 DEFINITION. Let $\mathbf{U}$ be a Grothendieck universe, an *U-set* is an element of $\mathbf{U}$. An *U-class* is a subset of $\mathbf{U}$. From now on, when we need a universe in our discussion, we will typically fix a universe $\mathbf{U}$ and use the following translated terminology, which we call the *conglomerate convention for $\mathbf{U}$* (**CCU**):

1. a *U-set* in the sense of **ZFC** is called a “set” in **CCU**;
2. a *U-class* in the sense of **ZFC** is called a “class” in **CCU**;
3. any set (not necessary elements or subsets of $\mathbf{U}$) in the sense of **ZFC** is called a “conglomerate” in **CCU**.

The concept “conglomerate” has been created to deal with “collections of classes”. As a result, in this terminology, we have the class $\mathbf{U}$ of all sets, and the conglomerate $\mathcal{P}(\mathbf{U})$ of all classes, but there is no “collection” of all conglomerates.

From now on, when we discuss categories, we will default to fixing an inaccessible cardinal, thereby giving a non-trivial universe $\mathbf{U}$. Categories will be understood as $\mathbf{U}$ categories by default. Under **CCU**, we can speak of “the category of all sets” and “the category of all classes”. When a larger category is needed, it suffices to consider a larger inaccessible cardinal.

In daily life and common mathematical theories, a single non-trivial universe is sufficient for carrying out rich investigations. However, this requirement is still quite demanding (since it assumes the existence of an inaccessible cardinal). In fact, the universe we more commonly encounter is the “ordinary mathematics” universe $\mathbf{V}_{\omega+\omega}$, which is a model of Zermelo set theory (**ZF** without the axiom of replacement). The reason is that $\mathbf{V}_{\omega+1}$ is sufficient for constructing the integers, $\mathbf{V}_{\omega+2}$ is sufficient for constructing the real numbers.

Categories

The above treatment of the details concerning categories is referred to as the set-theoretic foundation of category theory. At this point, we officially begin our theoretical study of category theory. Here are some common symbol simplifications.

$$\begin{aligned} gf &\text{ to mean } g \circ f \\ A \in \mathbf{C} &\text{ to mean } A \in \text{Ob}(\mathbf{C}) \\ f : A \rightarrow B \text{ or } A \xrightarrow{f} B &\text{ to mean } f \in \text{Hom}_{\mathbf{C}}(A, B) \end{aligned}$$

1.1.1 REMARK. We sometimes figuratively call morphisms arrows. The composition of morphisms corresponds to head-to-tail concatenation of arrows. Diagrams with arrows provide a convenient language for discussing categories. Among these, the most commonly used is the notion of a *commutative diagram*, where “commutative” means that different

compositions of arrows lead to the same result. For example, the commutativity of the diagram

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ x \downarrow & & \downarrow v \\ C & \xrightarrow{y} & D \end{array}$$

is equivalent respectively to $vu = yx$. The names of morphisms are often omitted from diagrams when they are self-evident or unimportant.

1.1.1 EXAMPLE. We could define a category **Pos** whose objects are all partially ordered sets $\langle P, \leq \rangle$. A morphism $\langle P, \leq \rangle \rightarrow \langle Q, \leq \rangle$ is an increasing function $f : P \rightarrow Q$, that is, for all $x, y \in P$, $x < y \rightarrow f(x) < f(y)$. One may check that **Pos** is indeed a category.

1.1.2 EXAMPLE. (See *Algebra: Chapter 0*, Paolo Aluffi, P20, Examples 3.3) Suppose that S is a set and $\sim$ is a relation on S satisfying the reflexive and transitive properties. Then we can encode this data into a category **C**:

- $\text{Ob}(\mathbf{C}) = \{x : x \in S\}$;
- $\text{Mor}(\mathbf{C}) = \bigcup_{a,b \in S} \text{Hom}_{\mathbf{C}}(a, b)$, where $\text{Hom}_{\mathbf{C}}(a, b) = \{(a, b) : a \sim b\}$.

Note that there are very “few” morphisms: at most one for any pair of objects. How to define the “composition” of two arrows ?

First, do we have “identities” ? If $a \in S$, we need to find an element $1_a \in \text{Hom}_{\mathbf{C}}(a, a)$. This is precisely why we are assuming that $\sim$ is reflexive: this tells us that $a \sim a$ for all $a \in S$, that is, $\text{Hom}_{\mathbf{C}}(a, a)$ consists of the single element (a, a) . We have no choice; we must let

$$1_a = (a, a) \in \text{Hom}_{\mathbf{C}}(a, a).$$

As for composition, let $a, b, c \in S$ and $f \in \text{Hom}_{\mathbf{C}}(a, b), g \in \text{Hom}_{\mathbf{C}}(b, c)$; we have to define a corresponding morphism $gf \in \text{Hom}_{\mathbf{C}}(a, c)$. Now, $f \in \text{Hom}_{\mathbf{C}}(a, b)$ tells us that $\text{Hom}_{\mathbf{C}}(a, b)$ is nonempty, which means $a \sim b$ and $f = (a, b)$. Similarly, $b \sim c$ and $g = (b, c)$. Now

$$a \sim b \wedge b \sim c \rightarrow a \sim c$$

since $\sim$ is transitive. This tells us that $\text{Hom}_{\mathbf{C}}(a, c)$ consists of the single element (a, c) . Thus we again have no choice: we must let

$$gf = (a, c) \in \text{Hom}_{\mathbf{C}}(a, c)$$

Is this composition associative ? If $f \in \text{Hom}_{\mathbf{C}}(a, b), g \in \text{Hom}_{\mathbf{C}}(b, c)$ and $h \in \text{Hom}_{\mathbf{C}}(c, d)$, then necessarily $f = (a, b), g = (b, c)$ and $h = (c, d)$, it follows that $gf = (a, c)$ and $hg = (b, d)$, and hence

$$h(gf) = (a, d) = (hg)f.$$

One last question, are these identities satisfy the identity law ? If $f \in \text{Hom}_{\mathbf{C}}(a, b)$, then $f1_a \in \text{Hom}_{\mathbf{C}}(a, b)$ and $1_b f \in \text{Hom}_{\mathbf{C}}(a, b)$, we have no choice that $f1_a = f$ and $1_b f = f$.

The most trivial instance of this construction is the category obtained from a set taken with the equivalence relation “=”, that is, the only morphisms are the identity morphisms: $\text{Hom}_{\mathbf{C}}(a, b) = \{(a, a) = 1_a : a \in S\}$ and $\text{Hom}_{\mathbf{C}}(a, b) = \emptyset$ if $a \neq b$. These categories are called the *discrete categories*.

1.1.3 EXAMPLE. Every category **C** has an *opposite category* $\mathbf{C}^{op}$, defined by “reversing the arrows”. Formally, $\text{Ob}(\mathbf{C}^{op}) = \text{Ob}(\mathbf{C})$ and $\text{Hom}_{\mathbf{C}^{op}}(X, Y) = \text{Hom}_{\mathbf{C}}(Y, X)$ for all objects X, Y . Identities in $\mathbf{C}^{op}$ are the same as in **C**; composition in $\mathbf{C}^{op}$ is the same as in **C** but with the arguments reversed: let $f \in \text{Hom}_{\mathbf{C}^{op}}(X, Y), g \in \text{Hom}_{\mathbf{C}^{op}}(Y, Z)$, define the composition of morphisms by $g \circ_{\mathbf{C}^{op}} f := f \circ_{\mathbf{C}} g$. One may check that it satisfies the associativity and identity law.

1.1.4 EXAMPLE. To give readers a sense of the extremely abstract nature of category theory, we now introduce a novel category. Let $\mathbf{C}$ be a category, and let $A \in \mathbf{C}$. We try to define a new category $\mathbf{C}_A$ whose objects are some morphisms in $\mathbf{C}$:

- $\text{Ob}(\mathbf{C}_A) = \{f \in \text{Hom}_{\mathbf{C}}(Z, A) : Z \in \mathbf{C}\}$
- $\text{Mor}(\mathbf{C}_A) = \bigcup_{f_1, f_2} \text{Hom}_{\mathbf{C}_A}(f_1, f_2)$, how to define the Hom-sets? For objects $f : Z_1 \rightarrow A$ and $f_2 : Z_2 \rightarrow A$, the Hom-set $\text{Hom}_{\mathbf{C}_A}(f, g)$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array}$$

For now, we won't rush to explain why it is defined this way. First, let us define a suitable composition so that it becomes a category. Let $f \in \mathbf{C}_A$, the identity $1_f \in \text{Hom}_{\mathbf{C}_A}(f, f)$ is the commutative diagram

$$\begin{array}{ccc} Z & \xrightarrow{1_Z} & Z \\ & \searrow f & \swarrow f \\ & A & \end{array}$$

Let $f_1 \in \text{Hom}(Z_1, A)$, $f_2 \in \text{Hom}(Z_2, A)$ and $f_3 \in \text{Hom}(Z_3, A)$. The two morphisms $f_1 \rightarrow f_2$ and $f_2 \rightarrow f_3$ in $\mathbf{C}_A$ correspond to putting the two commutative diagrams side-by-side:

$$\begin{array}{ccccc} Z_1 & \xrightarrow{\sigma} & Z_2 & \xrightarrow{\tau} & Z_3 \\ & \searrow f_1 & \downarrow f_2 & \swarrow f_3 & \\ & A & & & \end{array}$$

and then we define the composition of $f_1 \rightarrow f_2$ and $f_2 \rightarrow f_3$ by removing the central arrow, and we obtain the commutative diagram

$$\begin{array}{ccc} Z_2 & \xrightarrow{1_Z} & Z_3 \\ & \searrow f_2 & \swarrow f_3 \\ & A & \end{array} \circ \begin{array}{ccc} Z_1 & \xrightarrow{1_Z} & Z_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array} = \begin{array}{ccc} Z_1 & \xrightarrow{\tau\sigma} & Z_3 \\ & \searrow f_1 & \swarrow f_3 \\ & A & \end{array}$$

One may check that $\mathbf{C}_A$ is indeed a category. Categories constructed in this fashion are called the **slice categories** in the literature. Define a function $\Phi : \{\sigma \in \text{Hom}_{\mathbf{C}}(Z_1, Z_2) : f_1 \in \text{Hom}_{\mathbf{C}}(Z_1, A), f_2 \in \text{Hom}_{\mathbf{C}}(Z_2, A), f_2 = f_1\sigma\} \rightarrow \text{Hom}_{\mathbf{C}_A}(f_1, f_2)$ by $\Phi(\sigma) :=$

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ & \searrow f_1 & \swarrow f_2 \\ & A & \end{array}$$

It is clear that Φ is a bijection. In other words, each morphism in $\mathbf{C}_A$ uniquely determines a morphism in $\mathbf{C}$, and vice versa. For convenience, we simply denote the elements of $\text{Hom}_{\mathbf{C}_A}(f_1, f_2)$ as those morphisms in $\mathbf{C}$ that satisfy the commutative condition, i.e., $\text{Hom}_{\mathbf{C}_A}(f_1, f_2) = \{\sigma \in \text{Hom}_{\mathbf{C}}(Z_1, Z_2) : f_1 = f_2\sigma\}$.

1.1.5 EXAMPLE. We can similarly construct new categories in an analogous way. Let $\mathbf{C}$ be a category, and let $A \in \mathbf{C}$. We define a category $\mathbf{C}^A$:

- $\text{Ob}(\mathbf{C}^A) = \{f \in \text{Hom}_{\mathbf{C}}(A, Z) : Z \in \mathbf{C}\};$
- $\text{Mor}(\mathbf{C}^A) = \bigcup_{f_1, f_2} \text{Hom}_{\mathbf{C}^A}(f_1, f_2)$. The Hom set $\text{Hom}_{\mathbf{C}^A}(f_1, f_2)$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc}
 & A & \\
 f_1 \swarrow & & \searrow f_2 \\
 Z_1 & \xrightarrow{\sigma} & Z_2
 \end{array}$$

1.1.6 EXAMPLE. As an instance of a category in example 4. Let $\mathbf{C} = \mathbf{Set}$ and $A = \{*\}$. Call the resulting category $\mathbf{Set}^*$. An object in $\mathbf{Set}^*$ is a morphism $f : \{*\} \rightarrow S$ where S is any set in $\mathbf{Set}$. Clearly, f is determined by the element $s = f(*) \in S$. Thus, we may denote objects of $\mathbf{Set}^*$ as pairs (S, s) . A morphism between two such objects $(S, s) \rightarrow (T, t)$ corresponds then to a function $\sigma : S \rightarrow T$ such that $\sigma(s) = t$.

1.1.7 EXAMPLE. Similarly, we can further introduce analogous categories. Let $\mathbf{C}$ be a category, and given two objects $A, B \in \mathbf{C}$. We can define a new category $\mathbf{C}_{A,B}$:

- $\text{Ob}(\mathbf{C}_{A,B}) = \left\{ (f, g) := \begin{array}{ccc} & A & \\ f \swarrow & & \searrow g \\ Z & & B \end{array} : f \in \text{Hom}_{\mathbf{C}}(Z, A), g \in \text{Hom}_{\mathbf{C}}(Z, B) \right\}$
- $\text{Mor}(\mathbf{C}_{A,B}) = \bigcup_{(f_1, g_1), (f_2, g_2)} \text{Hom}_{\mathbf{C}_{A,B}}((f_1, g_1), (f_2, g_2))$. How to define the Hom-sets? For objects (f_1, g_1) and (f_2, g_2) , the Hom-set $\text{Hom}_{\mathbf{C}_{A,B}}((f_1, g_1), (f_2, g_2))$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc}
 & A & \\
 f_1 \swarrow & & \searrow f_2 \\
 Z_1 & \xrightarrow{\sigma} & Z_2 \\
 g_1 \swarrow & & \searrow g_2 \\
 & B &
 \end{array}
 \quad \text{or} \quad
 \begin{array}{ccc}
 & A & \\
 f_1 \swarrow & & \searrow f_2 \\
 Z_1 & \xrightarrow{\sigma} & Z_2 \\
 g_1 \swarrow & & \searrow g_2 \\
 & B &
 \end{array}$$

Since most of the descriptions in category theory are about morphisms, we define here some terminology concerning morphisms, whose definitions are obtained directly by generalizing the notions of bijection, surjection, and injection for functions between sets.

Unlike in set theory, where injectivity, surjectivity, and bijectivity are defined using elements, a general category may have objects that are not sets and morphisms that are not functions. Therefore, we must define the categorical analogues — monomorphisms, epimorphisms, and isomorphisms — using only the composition operation. Incidentally, the set-theoretic notions can be rephrased in terms of composition as well, confirming the consistency of the categorical definitions.

1.1.5 DEFINITION. Let $\mathbf{C}$ be a category. A morphism $f \in \text{Hom}_{\mathbf{C}}(A, B)$ is an *isomorphism* if it has a two-sided inverse under composition, that is, there exists $g \in \text{Hom}_{\mathbf{C}}(B, A)$ such that

$$gf = 1_A, \quad fg = 1_B.$$

1.1.3 PROPOSITION. (See *Algebra: Chapter 0*, Paolo Aluffi, P28, Proposition 4.2) *The inverse of an isomorphism is unique.*

Proof. Let $g_1, g_2 : B \rightarrow A$ act as the inverses of a given isomorphism $f : A \rightarrow B$. We need to verify that $g_1 = g_2$ in this case: $g_1 = g_1 1_B = g_1(fg_2) = (g_1f)g_2 = 1_A g_2 = g_2$, as desired. $\square$

Note that the argument really proves that if f is a morphism with a left-inverse g_1 and a right-inverse g_2 , then necessarily f is an isomorphism; $g_1 = g_2$; and this morphism is the unique inverse of f . Since the inverse of f is uniquely determined by f , there is no ambiguity in denoting it by f^{-1} .

1.1.4 PROPOSITION.

1. Each identity 1_A is an isomorphism and is its own inverse.
2. If f is an isomorphism, then f^{-1} is an isomorphism and further $(f^{-1})^{-1} = f$.
3. If $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}}(B, C)$ are isomorphisms, then the composition gf is an isomorphism and $(gf)^{-1} = f^{-1}g^{-1}$.

Two objects A, B of a category are *isomorphic* if there is an isomorphism $f : A \rightarrow B$. An immediate corollary of this proposition is that “isomorphism” is an equivalence relation. If two objects A, B are isomorphic, one write $A \cong B$.

1.1.8 EXAMPLE. Identities may be the only isomorphisms in a category: for example, this is the case in the category $\mathbf{C}$ obtained from the relation $\leq$ on $\mathbf{Z}$. Indeed, for $a, b \in \mathbf{C}$, there is a morphism $f : a \rightarrow b$ and a morphism $g : b \rightarrow a$ only if $a \leq b$ and $b \leq a$, that is, if $a = b$. So an isomorphism in $\mathbf{C}$ necessarily acts from an object a to itself; but in $\mathbf{C}$, there is only one such morphism, that is, 1_a .

1.1.9 EXAMPLE. On the other hand, there are categories in which every morphism is an isomorphism; such categories are called *groupoids*. See Example 3.1.2. If this relation also satisfies symmetry, i.e., it is an equivalence relation, then the category defined in the same way is a groupoid.

Below we define monomorphisms and epimorphisms, which correspond to injective and surjective maps in **Set**. The following definitions may not be what the reader expects, or they may raise some questions as to why they are defined this way:

1.1.6 DEFINITION. Let $\mathbf{C}$ be a category.

1. A morphism $f \in \text{Hom}(A, B)$ is a *monomorphism* if for all objects Z and all morphisms $g, h \in \text{Hom}(Z, A)$, $(fg = fh \iff g = h)$.
2. A morphism $f \in \text{Hom}(A, B)$ is a *epimorphism* if for all objects Z and all morphisms $g, h \in \text{Hom}(B, Z)$, $(gf = hf \iff g = h)$.

1.1.2 REMARK. Let $\mathbf{C}$ be a category and let $f : A \rightarrow B$ be an isomorphism. Then there exists $g : B \rightarrow A$ such that $gf = 1, fg = 1$. Clearly, f is a monomorphism: if $fh_1 = fh_2$, then $gf h_1 = gf h_2$ which implies $h_1 = h_2$. Similarly, f is also epimorphism. Does the converse also hold, as in the category of sets **Set**, where a bijection is equivalent to an injective and surjective map? This is clearly false. A simple example is the category of posets **Pos**, which we mentioned earlier. Let $f : a \rightarrow b$ be the unique morphism from a to b . It is a monomorphism because there is at most one morphism from any c to a , so the cancellation law holds trivially; similarly, it is an epimorphism. But f is clearly not an isomorphism. Later there will be another typical example: the category of rings **Ring** also has this property.

1.2 Universal Properties

The main purpose of category theory is to unify different branches of mathematics, which often share similar structures. For example, in set theory, the “product” generally refers to the Cartesian product; in point-set topology, the “product” refers to the product spaces; and in group theory, the “product” refers to the direct product of groups and so on. While the construction details of these “products” differ, they can all be defined by the universal property that we will study in this section. This is precisely the role of universal properties: they avoid discussing the specific details of what a product is in each context and instead focus on its “function” within the corresponding category.

1.2.1 DEFINITION. Let $\mathbf{C}$ be a category.

- We say that an object I of $\mathbf{C}$ is *initial* in $\mathbf{C}$ if for every object A of $\mathbf{C}$, there exists exactly one morphism $I \rightarrow A$ in $\mathbf{C}$, that is, $\text{Hom}_{\mathbf{C}}(I, A)$ is a singleton for every object A of $\mathbf{C}$.
- We say that an object F of $\mathbf{C}$ is *final* in $\mathbf{C}$ if for every object A of $\mathbf{C}$, there exists exactly one morphism $A \rightarrow F$ in $\mathbf{C}$, that is, $\text{Hom}_{\mathbf{C}}(A, F)$ is a singleton for every object A of $\mathbf{C}$.

1.2.1 EXAMPLE. A category need not have initial or final objects. Consider the special case of the category in example 1, where S is $\mathbf{Z}$ and the relation $\sim$ is $\leq$. This category has no initial or final object. Indeed, an initial object in this category would be an integer i such that $i \leq a$ for all $a \in \mathbf{Z}$, there is no such integer. Similarly, a final object would be an integer larger than every integer, and there is no such integer.

1.2.2 EXAMPLE. In $\mathbf{Set}$, the empty set $\emptyset$ is initial (the unique function from $\emptyset$ to $S \in \mathbf{Set}$ is the empty relation $f = \emptyset$). And $\emptyset$ is the unique initial object in $\mathbf{Set}$: taking any nonempty set A , then there is no function $A \rightarrow \emptyset$. Thus, A cannot be an initial object of $\mathbf{Set}$.

Moreover, $\mathbf{Set}$ also has final objects: for every set A , there is a unique function from A to a singleton $\{p\}$, that is the constant function. Thus, every singleton is final in $\mathbf{Set}$, and the final objects are not unique in this category.

1.2.1 THEOREM. Let $\mathbf{C}$ be a category.

1. If I_1, I_2 are both initial objects in $\mathbf{C}$, then $I_1 \cong I_2$.
2. If F_1, F_2 are both final objects in $\mathbf{C}$, then $F_1 \cong F_2$.

Proof. Since F_1, F_2 are final in $\mathbf{C}$, there exists a unique morphism $f : F_1 \rightarrow F_2$. Also there exists a unique morphism $g : F_2 \rightarrow F_1$. Consider $gf : F_1 \rightarrow F_1$, and this is clearly an identity 1_{F_1} by definition of $\text{Hom}(F_1, F_1)$. Similarly, $fg = 1_{F_2}$. The proof for initial objects is entirely analogous. $\square$

1.2.3 EXAMPLE. We emphasize again Example 2.2. Let 1 denote a singleton. (It does not matter what this element is called.) Then 1 has the property: for all sets $S \in \mathbf{Set}$, there exists a unique function from S to 1 (the constant function).

Phrases of the form “there exists a unique such-and-such satisfying some condition” are common in category theory. Properties such as this are called “universal” because they state how the object being described (in this case, the set 1) relates to the entire category which it lives (in this case, the set of all sets).

1.2.2 DEFINITION. Let $\mathbf{C}$ be a category, we say that an object $A \in \mathbf{C}$ is *universal with respect to some property* if A is terminal in $\mathbf{C}$.

For example 2.3, a singleton 1 is universal with respect to the property: for all $Z \in \mathbf{Set}$, there exists a unique function $Z \rightarrow 1$. And we call this property of “mapping from sets” informally. Similarly, $\emptyset$ is universal with respect to the property: for all $Z \in \mathbf{Set}$, there exists a unique function $\emptyset \rightarrow Z$, and we call this property of “mapping to sets”.

1.2.2 PROPOSITION. Consider the category $\mathbf{Set}$, let $A, B \in \mathbf{C}$. Then the product $A \times B$ satisfies the property: for all $Z \in \mathbf{Set}$ and two functions $f_A : Z \rightarrow A$ and $f_B : Z \rightarrow B$, there exists a unique function $\sigma : Z \rightarrow A \times B$ such that the diagram commutes (where π_A and π_B are the natural projections defined by $\pi_A(a, b) = a, \pi_B(a, b) = (b)$)

$$\begin{array}{ccccc}
 & & f_A & \rightarrow & A \\
 & \nearrow & & \searrow \pi_A & \\
 Z & \xrightarrow{\sigma} & A \times B & & \\
 & \searrow & & \nearrow \pi_B & \\
 & & f_B & \rightarrow & B
 \end{array}$$

Proof. To prove it, we define a function $\sigma : Z \rightarrow A \times B$ by $\forall z \in Z \sigma(z) = (f_A(z), f_B(z))$. This function makes the diagram commute: $\pi_A \sigma(z) = \pi_A(f_A(z), f_B(z)) = f_A(z)$, showing that $\pi_A \sigma = f_A$ and similarly $\pi_B \sigma = f_B$. Further, the definition is forced by the commutativity of the diagram; so σ is unique. $\square$

In fact, we may consider the proposition in the slice category $\mathbf{Set}_{A,B}$. Then the original proposition is restated as: for all $(Z, f_A, f_B) \in \mathbf{Set}_{A,B}$ there exists a unique commutative diagram in $\mathbf{Hom}((Z, f_A, f_B), (A \times B, \pi_A, \pi_B))$. And this proof tells us that in the category $\mathbf{Set}_{A,B}$, the object $(A \times B, \pi_A, \pi_B)$ is universal with respect to the property: for all $(Z, f_A, f_B) \in \mathbf{Set}_{A,B}$, there exists a unique function σ such that the diagram commutes

$$\begin{array}{ccc}
 & & \begin{array}{c} \xrightarrow{f_A} A \\ \nearrow \pi_A \end{array} \\
 Z & \xrightarrow{\sigma} & A \times B \\
 & & \begin{array}{c} \searrow \pi_B \\ \xrightarrow{f_B} B \end{array}
 \end{array}$$

Similarly, what is described here is actually that $(A \times B, \pi_A, \pi_B)$ is universal with respect to the property. Strictly speaking, if we want to define the product of two sets via a universal property, it is actually defined as $(A \times B, \pi_A, \pi_B)$, but defining it directly as $A \times B$ is unambiguous. In summary, we can use the universal property to define the product of a family of objects in a general category:

1.2.3 DEFINITION. Let $\mathbf{C}$ be a category and A and B objects of $\mathbf{C}$. A *product* of A and B is an object P of $\mathbf{C}$ together with two morphisms $\pi_A : P \rightarrow A$ and $\pi_B : P \rightarrow B$ satisfying the universal property: for all objects $Z \in \mathbf{C}$ and morphisms $f_A : Z \rightarrow A, f_B : Z \rightarrow B$, there exists a unique morphism $\sigma : Z \rightarrow P$ such that the diagram commutes:

$$\begin{array}{ccc}
 & & \begin{array}{c} \xrightarrow{f_A} A \\ \nearrow \pi_A \end{array} \\
 Z & \xrightarrow{\sigma} & P \\
 & & \begin{array}{c} \searrow \pi_B \\ \xrightarrow{f_B} B \end{array}
 \end{array}$$

This is the first concept we formally define using a universal property. It should be noted that concepts defined by universal properties are not unique. In many cases, different objects can satisfy the same universal property (for example, in $\mathbf{Set}$, both $(A \times B, \pi_A, \pi_B)$ and $(B \times A, \pi'_A, \pi'_B)$ satisfy the above universal property; they are both products of A and B). However, precisely because of this, they will simultaneously serve as final or initial objects in the corresponding category, and then by Theorem 3.2.1 they are isomorphic. In other words, they are “essentially” the same.

The following example illustrates well that products are not always of the form of a Cartesian product in different categories (in fact, some categories cannot even define a Cartesian product, such as those that do not include ordered pairs).

1.2.4 EXAMPLE. Consider the example 3.2.1. Does this category have products? Objects of this category are simply the integers $a, b \in \mathbf{Z}$. Call $a \times b$ the product of a and b . For all $z \in \mathbf{Z}$ and $f_a = (z, a), f_b = (z, b)$, that is $z \leq a$ and $z \leq b$, we have $z \leq \min\{a, b\}$, so there exists a morphism $\sigma = (z, \min\{a, b\})$ in $\mathbf{Hom}_{\mathbf{C}}(z, \min\{a, b\})$. Since σ is the only possible element in $\mathbf{Hom}_{\mathbf{C}}(z, \min\{a, b\})$ and $\pi_a \sigma = (z, a) = f_a$ and $\pi_b \sigma = (z, b) = f_b$, $\min\{a, b\}$ is the product of a and b .

1.2.4 DEFINITION. Let $\mathbf{C}$ be a category and A and B objects of $\mathbf{C}$. A *coproduct* of A and B is an object S of $\mathbf{C}$ together with two morphisms $\pi_A : A \rightarrow S$ and $\pi_B : B \rightarrow S$ satisfying the universal property: for all objects $Z \in \mathbf{C}$ and morphisms $f_A : A \rightarrow Z, f_B :$

$B \rightarrow Z$, there exists a unique morphism $\sigma : S \rightarrow Z$ such that the diagram commutes:

$$\begin{array}{ccccc} A & & f_A & & \\ & \searrow \iota_A & & \searrow & \\ & S & \xrightarrow{\sigma} & Z \\ & \nearrow \iota_B & & \nearrow & \\ B & & f_B & & \end{array}$$

1.2.3 PROPOSITION. *The disjoint union is a coproduct in $\mathbf{Set}$.*

Proof. Let A, B be any two sets in $\mathbf{Set}$, and let $A' = \{0\} \times A, B' = \{1\} \times B$. The disjoint union $A \coprod B$ of A and B is the set $A' \cup B' = (\{0\} \times A) \cup (\{1\} \times B)$. And the functions ι_A, ι_B are defined by $\iota_A(a) = (0, a), \iota_B(b) = (1, b)$. Now, let $f_A : A \rightarrow Z, f_B : B \rightarrow Z$ be arbitrary morphisms. If the diagram commutes:

$$\begin{array}{ccccc} A & & f_A & & \\ & \searrow \iota_A & & \searrow & \\ & A \coprod B & \xrightarrow{\sigma} & Z \\ & \nearrow \iota_B & & \nearrow & \\ B & & f_B & & \end{array}$$

then the function $\sigma : A \coprod B \rightarrow Z$ would satisfy $\sigma(c) = f_A(a)$ if $c = (0, a) \in A'$ and $\sigma(c) = f_B(b)$ if $c = (1, b) \in \{1\} \times B$. This definition is forced upon the commutativity, proving that σ is unique. $\square$

This proposition also illustrates once again that, in the same category, there may be different objects satisfying the same universal property. Here, as long as the two defined sets A' and B' are disjoint, they both satisfy the above universal property, and therefore they are isomorphic.

1.2.4 PROPOSITION. *Consider the category $\mathbf{Set}$, let $\sim$ be an equivalence relation defined on a set A . Then the quotient $A/\sim$ satisfies the property: for all $Z \in \mathbf{Set}$ and the function $\varphi : A \rightarrow Z$ satisfying $(a_1 \sim a_2) \rightarrow \varphi(a_1) = \varphi(a_2)$, there exists a unique function $\bar{\varphi} : A/\sim \rightarrow Z$ such that the diagram commutes (where $\pi : A \rightarrow A/\sim$ is the canonical projection defined by $\pi(a) = [a]_\sim$, so that $(a_1 \sim a_2) \rightarrow \pi(a_1) = \pi(a_2)$):*

$$\begin{array}{ccc} A/\sim & \xrightarrow{\bar{\varphi}} & Z \\ & \nwarrow \pi \quad \nearrow \varphi & \\ & A & \end{array}$$

Proof. To prove it, let $[a]_\sim$ be an arbitrary element of $A/\sim$. If the diagram is commutative, then necessarily

$$\bar{\varphi}([a]_\sim) = \varphi(a),$$

this tells us that $\bar{\varphi}$ is indeed unique if it exists. Hence we have to check that $\bar{\varphi}$ is well-defined, that is, if $[a_1]_\sim = [a_2]_\sim$, then $\varphi(a_1) = \varphi(a_2)$; and indeed

$$[a_1]_\sim = [a_2]_\sim \rightarrow a_1 \sim a_2 \rightarrow \varphi(a_1) = \varphi(a_2).$$

$\square$

In fact, we may consider the proposition in a category $\mathbf{C}$, in which $\text{Ob}(\mathbf{C}) = \{f \in \text{Hom}_{\mathbf{C}}(A, Z) : Z \in \mathbf{C} \wedge (a_1 \sim a_2 \rightarrow f(a_1) = f(a_2))\}$, and $\text{Mor}(\mathbf{C}) = \bigcup_{f_1, f_2} \text{Hom}_{\mathbf{C}}(f_1, f_2)$, the Hom-set $\text{Hom}_{\mathbf{C}}(f_1, f_2)$ consists of all the commutative diagrams of the form

$$\begin{array}{ccc} Z_1 & \xrightarrow{\sigma} & Z_2 \\ & \nwarrow f_1 \quad \nearrow f_2 & \\ & A & \end{array}$$

Then the original proposition is restated as: for all $\varphi \in \mathbf{C}$, there exists a unique commutative diagram in $\text{Hom}_{\mathbf{C}}(\pi, \varphi)$. From this proof, we have the conclusion that, in the category $\mathbf{C}$ defined above, the object $(A/\sim, \pi)$ (ordered pairs are used to denote objects here to emphasize the quotient set) is universal with respect to the property: for all $(Z, f) \in \mathbf{C}$, there exists a unique function $\bar{\varphi}$ such that the diagram commutes:

$$\begin{array}{ccc} A/\sim & \xrightarrow{\exists! \bar{\varphi}} & Z \\ & \swarrow \pi \quad \searrow \varphi & \\ & A & \end{array}$$

In fact, what is described here is actually that $(A/\sim, \pi)$ is universal with respect to the property. Strictly speaking, if we want to define the quotient set of a set via a universal property, it is actually defined as $(A/\sim, \pi)$, but usually we directly call it $A/\sim$ without ambiguity.

One could attempt to define quotients in a general category using universal properties, but that would require heavy abstraction. Since we prefer to think of quotients via equivalence relations, we shall focus on categories that are closely related to sets — namely, **concrete categories**.

1.2.5 DEFINITION. A *concrete category* is a category $\mathbf{C}$ together with a function σ that assigns to each object A of $\mathbf{C}$ a set $\sigma(A)$ called the *underlying set of A* in such a way that

1. every morphism $A \rightarrow B$ of $\mathbf{C}$ is a function on the underlying sets $\sigma(A) \rightarrow \sigma(B)$;
2. the identity morphism 1_A of $\mathbf{C}$ is the identity function on the underlying set $\sigma(A)$;
3. composition of morphisms in $\mathbf{C}$ agrees with the composition of functions on the underlying sets.

1.2.5 EXAMPLE. Consider the category $\mathbf{Pos}$ of all the partially ordered sets. Define the function $\sigma : \text{Ob}(\mathbf{Pos}) \rightarrow \text{Ob}(\mathbf{Set})$ by $\sigma(\langle P, \leq_P \rangle) = P$. One may check that the category $\mathbf{Pos}$ together with the function σ is indeed a concrete category.

1.2.6 DEFINITION. Let $\mathbf{C}$ be a concrete category and A an object of $\mathbf{C}$. We can talk about $P \subset \mathcal{P}(\sigma(A))$, a set of nonempty disjoint subsets of $\sigma(A)$. A *quotient* of A by P is the object $\bar{A}$ in $\mathbf{C}$ endowed with a morphism $\pi : A \rightarrow \bar{A}$ which satisfies $\pi(a) = \pi(b)$ for all $S \in P$ and $a, b \in S$ satisfying the universal property: for all $Z \in \mathbf{C}$ and morphisms $\varphi : A \rightarrow Z$ such that for all $S \in P$, $\varphi(a) = \varphi(b)$ for all $a, b \in S$, there exists a unique morphism $\bar{\varphi} : \bar{A} \rightarrow Z$ such that the diagram commutes:

$$\begin{array}{ccc} \bar{A} & \xrightarrow{\exists! \bar{\varphi}} & Z \\ & \swarrow \pi \quad \searrow \varphi & \\ & A & \end{array}$$

Concrete categories are useful since one has available not only the properties of a category, but also certain properties of sets, subsets, etc. Since in virtually every concrete category we are interested in, the function σ assigns to an object its underlying set in the usual sense (as in the examples above), we shall denote both the object and its underlying set by the same symbol and omit any explicit reference to σ .

2

Category Theory II

2.1 Functors

2.2 Limits

A Equaliser and Kernel